

ATISHAY JAIN

+1 (401) 919-9415 [◇ atishay_jain@brown.edu](mailto:atishay_jain@brown.edu) [◇ atishayjain-aj.github.io](https://github.com/atishayjain-aj)

[LinkedIn](#)

[GitHub](#)

[Google Scholar](#)

EDUCATION

Brown University
Ph.D., Computer Science

September 2021 - Present
GPA: **4/4**

IIT (Indian Institute of Technology), Gandhinagar
B.Tech, Computer Science and Engineering | President's Gold Medalist

August 2016 - July 2020
GPA: **10/10**

SKILLS

Languages: Python, R, C, Java, C++

Libraries: PyTorch, PyTorch Geometric, Keras, Sklearn, NLTK, Spacy, Tensorflow

GRADUATE RESEARCH PROJECTS (BROWN UNIVERSITY)

Zero-Shot Enhancement of Foundation Models via Graph Diffusion; [August 2025 - Current]

- Designed a graph diffusion framework to integrate spatial information in spatial transcriptomics data to improve the performance of existing foundation models with no extra training
- Conducted extensive comparative analysis against multiple foundation models and demonstrated superior performance consistently, even in comparison to specialized methods
- [Submitted](#) to RECOMB 2026

Nested Graph Neural Networks for Improved Spatial Clustering; Prof. Ritambhara Singh; [August 2023 - August 2025]

- Implemented a nested Graph Neural Network (GNN) using PyTorch Geometric, designing a custom architecture for unsupervised clustering of spatial transcriptomics data to enable cell annotation and understand disease pathology
- Validated model performance through rigorous benchmarking against state-of-the-art techniques, GraphST and MuCoST, achieving superior cluster separation
- [Published](#) peer-reviewed research at ACM-BCB 2025 and presented posters at Intelligent Systems for Molecular Biology (ISMB), Montreal 2024 and Cold Spring Harbor Laboratory, 2024 New York, demonstrating strong technical communication and documentation skills

Resource-Efficient Semantic Segmentation via Graph Neural Networks; Prof. Ritambhara Singh; [August 2021 - August 2023]

- Developed a graph neural network-based framework using PytorchGeometric and Sklearn to perform memory-efficient semantic segmentation on large microscopy images
- Reduced computational overhead by 10x–1000x (1–3 orders of magnitude) compared to standard U-Nets, while maintaining competitive pixel-level accuracy benchmarks
- [Published](#) in Microscopy journal and presented at Cold Spring Harbor Laboratory, 2022 New York

Comparative Evaluation Framework for Data-Upscaling Architectures; Prof. Ritambhara Singh; [August 2020 - August 2022]

- Surveyed and reviewed multiple state-of-the-art models that upscale Hi-C data to reduce sequencing costs
- Conducted rigorous performance testing across heterogeneous datasets using multiple quantitative metrics to identify advantages and disadvantages of the models and methods used to train them

- [Published](#) findings in a peer-reviewed publication in the Genes journal

INTERNSHIPS

Caltech (California Institute of Technology); Prof. Mani Chandy; [May 2019 - July 2019]

- Developed algorithms in Python, for Caltech's IoTpy framework (which enables development of applications based on streaming data such as sensors, website/network monitoring, web-clicks, and audio signals)
- Programmed heavy hitter algorithms such as Misra Gries, Count-Min Sketch and Count Sketch, as well as an online version of Principal Component Analysis (PCA), a dimensionality reduction algorithm
- Created audio processing libraries for real time modifications such as reverberations and pitch shifting

IIT Gandhinagar; Prof. Anand Sengupta; [May 2018 - December 2019]

- Designed machine learning models, using Keras and PyTorch, to de-noise gravitational waves, allowing researchers to observe astronomical events such as binary black hole mergers
- Analysed neural network models, which use Recurrent Neural Networks (RNNs), Long Short Term Memory (LSTM) Cells with attention, Auto-Encoders, and Generative Adversarial Networks (GANs) to rapidly de-noise gravitational waves of varying signal-to-noise ratios

Hewlett Packard Enterprise; Mr. Suhas Shivanna; [June 2017 - July 2017]

- Programmed a Generic Data Mining Domain Modeller using HTML and JavaScript, which would make it easy to collect telemetry data from enterprise network equipment for further analysis
- Researched and presented highlights of Quantum Key Distribution to the security project team

UNDERGRADUATE RESEARCH PROJECTS (IIT GANDHINAGAR)

SVD Decomposition to Identify Genes for Cancer Classification; Prof. Anirban Dasgupta; [August 2019 - December 2019]

- Implemented and tested a singular value decomposition based algorithm on gene expression data to select genes for breast and lung cancer classification
- Collaborated with Zydus Hospital, Ahmedabad for extending the solution to other cancer types

Feature Hashing and Fairness; Prof. Anirban Dasgupta; [January 2019 - April 2019]

- Analyzed the effect of feature sketching on fairness using Python
- Inspected change in accuracy and fairness (equal odds and equal opportunity) of a Support-Vector Machine (SVM) model, after hashing the input data

Defending Neural Networks Against Adversarial Attacks; Prof. Nipun Batra; [January 2019 - April 2019]

- Explored and adapted (using Python and PyTorch) state-of-the-art defenses such as ensembles, defensive distillation, and Defense-GAN against adversarial attacks (FGSM attacks)
- Benchmarked adapted models' performance against Defense GAN model in protecting a neural network

Detecting Insults in Social Commentary; Prof. Mayank Singh; [August 2018 - December 2018]

- Identified offensive comments on social media
- Used Python libraries NLTK and ScikitLearn for text pre-processing (Natural Language Processing), and machine learning classifiers such as logistic regression, neural networks and SVMs for identification

PUBLICATIONS

- **Jain A.**, Pham, T.M., Laidlaw D.H., Ma Y., and Singh R. Diffusion-based Representation Integration for Foundation Models Improves Spatial Transcriptomics Analysis. Submitted to *Research in Computational Molecular Biology (RECOMB)*, May 2026 [[conference_link](#)]
- **Jain A.**, Laidlaw D.H., Ma Y., and Singh R. Improved Spatial Transcriptomics Clustering with Nested Graph Neural Networks. *ACM Conference on Bioinformatics, Computational Biology, and Health Informatics (ACM-BCB)*, October 2025 [[link](#)][[conference_link](#)]
- **Jain A.**, Laidlaw D.H., Bajcsy P., and Singh R. Memory-efficient semantic segmentation of large microscopy images using graph-based neural networks. *Microscopy*, Oct 2023 [[link](#)] [[journal_link](#)]
- Murtaza G., **Jain A.**, Hughes M., Wagner J., and Singh R. A Comprehensive Evaluation of Generalizability of Deep Learning-Based Hi-C Resolution Improvement Methods. *Genes*, Dec 2023 [[link](#)] [[journal_link](#)]
- Gohil V.*, Narayanan S.D.*, and **Jain A***. [Re] One ticket to win them all: generalizing lottery ticket initializations across datasets and optimizers. *ReScience C* 6, 2, #4, Feb 2020 [[link](#)] [[journal_link](#)]
- Dutta R.*, Gohil V.*, and **Jain A***. Effect of Feature Hashing on Fair Classification. *ACM India Joint International Conference on Data Science and Management of Data (CoDS-COMAD)*, Hyderabad, India, Jan 2020 [[link](#)] [[conference_link](#)]

INVITED TALKS AND POSTERS

Talk at ACM-BCB 2025, Philadelphia - Clustering Spatial Transcriptomics Data with Nested Graph Neural Networks

Lightning Talk and Poster at CSHL Biological Data Science Meeting 2024, New York - Clustering Spatial Transcriptomics Data with Nested Graph Neural Networks

Poster at Intelligent Systems for Molecular Biology 2024, Montreal - Clustering Spatial Transcriptomics Data with Nested Graph Neural Networks

Talk at CSHL Biological Data Science Meeting 2022, New York - Scalable and memory efficient segmentation of large microscopy images using graph-based neural networks

Talk at PyData Gandhinagar 2018 - Applications of Machine Learning Across Domains

TEACHING ASSISTANT EXPERIENCE

Deep Learning in Genomics (Brown University)

Fall 2023

Writing Lab (IIT Gandhinagar)

Spring 2019

Computing (Python) (IIT Gandhinagar)

Fall 2017, Fall 2018

HONORS AND RECOGNITION

- Awarded the [President's Gold Medal](#) for obtaining the highest GPA among all B.Tech students graduating in 2020 from IIT Gandhinagar.
- Awarded the [Scholarship for Academic Excellence](#) for obtaining first rank in class, IIT Gandhinagar for the years 2018-19, 2019-20
- Appeared on the [Dean's list](#) (higher than 9.0 Term GPA) for all semesters at IIT Gandhinagar
- Placed in the top 0.2% of IIT-JEE (Joint Entrance Examination), 2016 with over 1.1 million candidates
- Won [HP Code Wars](#), Bangalore, 2015 - a prestigious high-school coding competition conducted by Hewlett Packard where approximately 100 teams participated

SERVICES

- Sub-reviewer for Nature, ICML, NeurIPS, ICLR, and RECOMB conferences.
- Co-founded [Gandhinagar chapter of PyData](#) and organised 4 meetups to expand membership over 1000